# KANAK CHAUDHARI

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## PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building the internal tools, services, and automation that engineering teams rely on: deployment and release pipelines, build tooling, log collection, observability, and access management. Writes production software in Go and Python (plus Ruby and C) for containerized environments, and works across the stack from application code down to networking and databases. Open-source contributor and conference speaker.

## TECHNICAL SKILLS

**Programming Languages:** Go, Python, Ruby, C, JavaScript, Bash, PowerShell
**Platform and Developer Tooling:** CI/CD (Jenkins), build and release pipelines, Docker, Kubernetes, Docker Swarm, Ansible
**Observability and Log Collection:** ELK, Elastic Agent/Fleet, APM, Prometheus, Grafana, Pyroscope
**Frameworks:** Ruby on Rails, Django, FastAPI, Flask, gRPC
**Data and Messaging:** MySQL, PostgreSQL, Redis, RabbitMQ
**Cloud and Security:** AWS, DigitalOcean, Vault PKI, WireGuard, OAuth2, eBPF, Nginx
**Languages Spoken:** English (business), Japanese (approx. JLPT N2 level), Hindi

## PROFESSIONAL EXPERIENCE

**Full Stack Software and Infrastructure Developer — PriceLabs** **09/2023 - Current**

**– Internal Tools and Services:**
- Designed (full PRD) and built Underlease, an internal cloud resource management platform in Go, HTMX, and SQLite that tracks resource lifecycle and cleanup for engineering teams.
- Built an automated offboarding service that revokes VPN, admin panel, and jump server access when an account is deactivated, eliminating stale credentials and manual access cleanup.
- Designed and open-sourced a Redis-backed rate limiter with Fixed and Rolling Window algorithms.
- Created a command-driven job system using Redis, Sidekiq, and Rails, reducing system load by 5x.
- Built an internal TLS spoofing proxy service processing millions of requests per day.
- Migrated an internal Python MCP server from a single process to gunicorn with uvicorn workers, including a full security and architecture audit.
- Built a centralized audit and monitoring system for the production jump server, giving full traceability of administrative access and privileged operations.

**– CI/CD, Build, and Release Workflows:**
- Created Jenkins deployment pipelines supporting staggered releases and rollback procedures.
- Introduced and built an SCA pipeline to detect vulnerable dependencies before release.
- Led containerization of the legacy push pricing system, orchestrated with Docker Swarm.
- Architected an Infrastructure-as-Code solution in Ansible that deploys services, scripts, and monitoring across distributed environments, achieving 99.9% uptime.
- Developed a cost-efficient database backup pipeline using XtraBackup and Docker.

**– Log Collection and Observability:**
- Implemented an ELK-based logging platform with monitoring and failover, and extended log collection across the server fleet with Elastic Agent/Fleet.
- Built a two-tier Vault PKI certificate hierarchy with Vault Agent automation to issue and rotate certificates for the log collection fleet.
- Set up an APM server from scratch and tuned clients for reliable tracing under high load.
- Deployed Pyroscope continuous profiling fleet-wide, tracing about 78% of Sidekiq CPU to the logging pipeline.
- Wrote a custom Prometheus exporter and watchdog services, and built PMM dashboards and Grafana alerts for critical systems.

**– Backend and Reliability Engineering:**

- Architected and ran the RabbitMQ migration from a single node to an HA multi-node quorum cluster, driving application code changes, documentation, and load testing.
- Planned and implemented a Redis Sentinel architecture, migrating from a single-node setup.
- Architected scraping infrastructure handling 13 million+ listings, and made scraping 6x faster by reducing database connections and adding prepared statements with batch insertions.
- Improved API response times by cutting authentication database calls and moving session management to Redis.
- Redesigned billing job execution into smaller controlled batches, reducing peak memory by 30%; reduced Sidekiq resource consumption through Ruby memory management tuning.
- Managed and upgraded the MySQL cluster, resolved deadlocks, and ran database security and performance audits.
- Migrated Layer-7 traffic to HTTP/2 and built a WireGuard VPN for network isolation.

**DevOps and Security Developer — Data Cloud Gaze** **10/2024 - 02/2025**

- Created a PostgreSQL containerization tool with plugin extensibility, simplifying and accelerating database deployment for developers.
- Architected and implemented cross-platform CI/CD pipelines with multi-cloud storage integration.
- Wrote build pipelines and scripts in Bash and PowerShell for Linux and Windows targets.
- Designed and built a license server for secure software distribution.
- Engineered DLL protection mechanisms to harden applications against reverse engineering.

**Software Engineer — RyoCurrency** **10/2021 - 10/2022**

- Architected and deployed a full-stack application from initial planning to production, with a REST API in Django and PostgreSQL handling millions of records.
- Containerized the entire stack (Django backend, scrapers, and Nginx) with Docker.
- Built a real-time visualization system for Apache logs using a Sanic server with Nginx integration.
- Optimized search with the Manticore search engine, achieving a 200% performance improvement.
- Engineered a Man-in-the-Middle proxy system to analyze and decode mobile application traffic.

## PROJECTS AND CONTRIBUTIONS

**Omamori — Enterprise VPN and Secure Access Platform (500+ active users)** omamorisec.net

- Designed and built a production VPN platform in Go, with gRPC between services and a clear split between control plane (authentication, policy, billing, provisioning) and WireGuard-based data plane.
- Implemented multi-tenant user and profile management with per-organization policy enforcement, distributed rate limiting, and profile-level DNS filtering.
- Built dynamic WireGuard configuration provisioning and automated key lifecycle management.
- Deployed AWS infrastructure (IAM, EC2, KMS, S3, private VPC routing) with telemetry and observability pipelines.
- Integrated Chargebee subscription billing with idempotent webhook processors and a state synchronization layer that keeps feature gating consistent with billing events.

**ShadowStage — eBPF-based TLS Capture Tool**

- Building a TLS traffic capture tool in Go with cilium/ebpf, using uprobes on Go crypto/tls and OpenSSL.

## ACHIEVEMENTS

- Winner of the All-India Hackathon in Mumbai among 336 participants (2023)
- Semi-Finalist at the All-India Hackathon at Charotar University
- Speaker at MumPy 2024: “Secure Distribution Strategy for Commercial Applications”

## EDUCATION

**Bachelor of Technology in Information Technology** **GPA:** 9.0/10.0
Savitribai Phule University (Pune University)